Car Design Final Report
ME 2160 – Intro to Mechanical Engineering Design
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Contributions:

Valeria: Wrote the Abstract, Introduction, Design Description, Results and Conclusion. Did the Drawings and exploded views. Both Partners wrote the design description and the SolidWorks parts were split evenly between both members.

Mateo: Wrote the Background Research, Goal Statement, Task Specifications and Design Description. Wrote the design description alongside partner. Did the rendering and Animation. Both partners helped with the assembly and construction of the project.

In a team of two, we underwent the construction of a small battery powered plywood car that would travel 10 feet. Our team was inspired by the 2025 McClaren Formula One car and underwent the process of attempting to create a car with some of its features. With the freedom to create any transmission, our group decided to try out a 5:1 gear ratio transmission that would be powered by 2 AA batteries. We created four different iterations with the first two being used to test the transmission of the car and the fourth one unfortunately not working mechanically. Throughout the different iterations different issues such as friction, excess weight and misalignment in the gears caused large changes to the design of the car. The first iteration was built from cardboard which helped understand the build of the car and how to best design it on SolidWorks. We then created a rectangular car out of laser cut pieces in which the 5:1 gear ratio was tested. Upon discovering a problem with the friction provided onto the shaft from the frame, we changed the transmission into a 9:1 gear ratio. The second iteration was created to accommodate the change in gear ration and the third iteration allowed for the inspiration from the F1 car to be combined with the transmission. The fourth iteration drew a lot of aesthetics from F1 but did not function due to excess weight. The third iteration became the final design used and was adjusted to improve its performance. Throughout the process we made changes such as cutting off excess wood, changing the frame and improving the transmission. We solved issues with the construction such as switching from wood glue to hot glue for better control and electrical issues such as the battery holder melting due to the heat of the batteries which lead to a switch in wires and motor. The final design was a laser-cut plywood frame with a simple clustered transmission, single tapered top, and an integrated power switch. It managed to successfully travel 10ft in 1.97 s along a straight path and had a mass of 123.4 g. This car showed improvement from the other heavier iterations and an improvement in the stability and functionality of the design. This result shows how consideration of torque, friction, weight and the willingness to compromise aesthetic goals for functionality leads to a strong car. This project challenged and grew our understanding of engineering and its principles and helped us practice skills which will be helpful in other engineering avenues.

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Introduction:

Our team was tasked to create car parts and a transmission to create a car that could successfully travel 10ft unassisted. Other than a couple general guidelines our team had full creative liberty and decided to take inspiration from what we consider the best type of motorsport, Formula One, our team decided to take inspiration from the 2025 World Constructors Champion the Formula One 2025 McClaren car. This design is very complex and warranted various iterations of the car to find a car design that fit our aesthetic goals and completed the 10ft. Through the various iterations our group created, we had to find creative methods of solving the problems and difficulties we faced. We faced limitations due to our resources and time constraint, so we had to compromise our original plans and be comfortable making big changes to our car. Throughout our iterations we created very different cars, we first began this project by creating a very basic car model that would allow us to try out different transmissions and learn more about what was possible. Once this car was tested through the proof of concept, we did various attempts to create the car with the aesthetics and speed we wanted. This project proved to be difficult but very rewarding process as a lot was learned and although our plans changed throughout the design process, we still created something to be proud of.

Background Research

To begin the process of creating the car, we first had to research transmissions and find one that would work for our car. Transmissions themselves add no power to the car and instead change the power from the power unit to fit the needs of car or mechanism it is in. For this project it was recommended we use a 3:1 ratio which indicates that the motor would rotate three times for every one rotation of the shaft the wheels are on. This ratio is determined by the gears that connect the motor and shaft. A higher torque output for a transmission is needed to make cars go faster since it makes the wheels rotate faster and with more strength. For F1 cars we found the torque produced by the transmission isn't very high since the weight of the car is very low so higher acceleration and speed is easier to achieve.

Finding out that F1 cars are very light, which is what allows their quick movement, we began researching how to reduce the weight of our car. The main method found was to change the material of the car, for this project however that was not possible, so we had to find other methods to reduce the weight. We found that using less wood by any means would help reduce the weight, which could be done by removing parts of wood such as the top of the car or drilling speed holes. Our group wanted to use the speed holes as they would help give our car a more unique design and would also remove the weight with a little more control. We also looked at how to reduce the wood used via increasing the aerodynamics of the car since it would require less wood in the front. F1 cars are modeled with a very thin and long nose which is used to break through the air faster since there is less surface area. As well a front and back wing as they provide down force which makes the car be closer to the ground reducing it surface area once more. By removing some materials from the front and readding them in a way that benefits the car, the mass stays relatively the same but it improves the way the car behaves.

This project also allowed for multiple power units to be used. My group considered adding an additional set of batteries and or motor to provide more power for the car. When researching how to do this we stumbled across the Honda Two Motor Hybrid Electric Power Train, which uses two motors and proved to be very effective in increasing the top speed and helping the cars acceleration. The way the car is built, has the motors parallel to each other which simplifies the connections to the gears and thus improves the torque output of the transmission without complicating the system. We also looked at F1 car engines which have very strong power units and pair it with two electric motors in a smaller cluster in the back of the car. These two motors could feed either the same shaft or provide torque to either shaft which changes the car from a back wheel to four-wheel drive.

Goal Statement

Our team set out to create a car akin to a Formula One car. Based off our previous research we valued some things above the rest those being, a high gear ratio to off put the weight of the car, reducing the friction on the shaft to not have to add another power sources/motor and less surface area as it would reduce weight and add some aerodynamics to our car. The frame would feature a front and rear wing with a tapered frame to make the nose of the car long, thin and narrow in the front. The front and rear wing would be made from the plywood and would add some downforce to the car as the air would go up the wing. The shaft at the front of the car would be slightly shorter than the back shaft to create less surface area in the front of the car and thus increase the aerodynamics of the car and reduce some weight. We would connect the frames by making them interlock to have a stable frame that would be resilient during testing. The motor would be composed of two gears, and we were aiming for a 5:1 or higher ratio. The motor would go along the thicker part of the car in the back with it spreading out into the thinner parts of the car. There would be a switch placed on top to turn the car on and off via the wires connected to the two AA batteries and the motor. We also decided on just back wheel drive as that orientation and size for the transmission fit with the rest of our design the best. The shafts would connect through the car via small holes cut out in the frame which would hold up the frame of the car and the amount of contact between the shafts and the frame would be reduced when possible.

Task Specifications

For our task in this project, we defined as anything that we would need to keep our car from malfunctioning. This mainly included keeping the car moving in a straight line and going the 10 feet. Tasks that would uphold the car to this standard included a fixed stepper motor, a switch to stop the car, a transmission that does not have the gears misaligned (which reenforces the idea of a fixed stepper motor), and axles that do not cause the car to turn. Additionally, after the first complete car, we then decided on the final list of tasks for this project as: **optimizing weight, reducing friction, aligning our transmission, a strong frame to hold the motor and gears, a taper profile(for more of an aerodynamic shape similar to other cars), motor and transmission in the rear of the car, and laser cutting holes in plywood for axles.** Our team will need to have completed these in order to make sure our car is functional consistently.

Design Description

Once assigned the task, the very first objective was to find a transmission orientation and torque output that fit our wants and needs. We wanted to see how the transmission would be set up in the actual form of the car, therefore we took one 10 tooth gear on the motor and a 32-tooth gear which fulfilled the 3:1 gear ratio requirement, this created a very simple transmission and decided to build a car around it. As seen in Figure 1 we made this iteration out of cardboard in a tapered base using plastic frames to connect the shafts and wheels to the frame to gauge how the actual transmission would sit in the car and what parts we would need to create to make the transmission stable. This iteration did not run but also helped create the first laser cut design for the car as it showed how large the pieces would be and if the envisioned transmission would fit into the design we had planned. This model also helped us to see what parts would be needed that we hadn't considered before. As we discovered that we would need a gear box or a way of holding the motor and batteries into place so they wouldn't move and break apart while running.

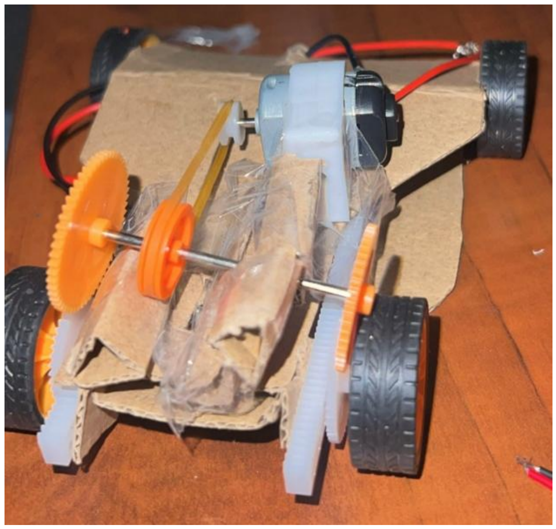


Figure 1: image of cardboard-frame car

After the first iteration was used to visualize the car, the actual measurements of the components were taken to begin the computer design process of our actual pieces. The point of this model was not to be functional but get rough estimates of what the car needs to appear like. The measurements for the components were taken but we didn't apply tolerance to them. To test out our first actual car we decided to create a basic rectangular frame in CAD and to cut it out to see how the transmission we had created would function, this iteration is seen in Figure 2. One of our task specifications was to reduce as much friction as possible on the shafts due to understanding that more friction on the shafts meant we would need a stronger torque output to counteract the friction, which could indicate having to complicate or change the transmission. To reduce the friction on the shafts, we decided to make the holes on the frame slightly larger by .005in so the frame would still be held up but the holes wouldn't be so exact that it would provide contact on all sides of the shaft. When this car went to be tested, we found that the car worked fine when it was being held up but once placed on the ground the car would not work. Upon inspection we found that the gears were too far apart, although they were in contact with each other, they had a very weak connection. This resulted in the rotation from the gear connected to the motor to not fully engage and transmit the torque into the gear connected to the shaft. This made it so the car would not run when the car was placed on the ground since the frame added friction to the shaft due to gravity. This made it so the weak energy being fed into the shaft was not enough to counteract the static friction and get the car to move. We fixed this by removing the piece we had created as a gear box as we needed to move the motor closer to the shaft to increase the contact between the two gears. We were unable to include that piece of the gear box to this iteration because there wasn't enough room between the battery holder, motor and shaft to add it back in. We also wanted to include a secure method of combining the sides, bottom and top of the frame instead of just gluing them plainly together. The pieces were created using slight cuts outs that match the other pieces so they would interlock and when glued together would lie flat against each other.

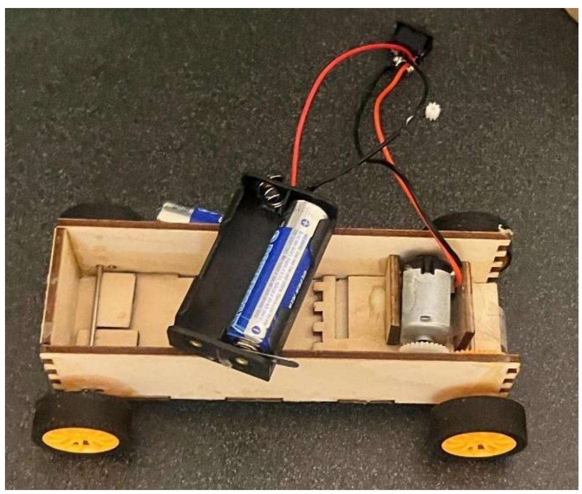


Figure 2: 2nd iteration, lasercut wooden frame car

Once this iteration was done, our team began to think about the various ways in which we could incorporate our initial design goal into the car model and transmission we had created. The first part of the design that we wanted to incorporate would be the tapered frame which was thicker in the back and got narrower at the front of the car. However, upon seeing our transmission, we measured the width of the transmission and there wouldn't be a way to make the car get substantially thinner at the front, in a way that fit our aesthetic goals while keeping the same transmission layout. We decided to keep the transmission as its size was already reduced and simplified, and it sat at a comfortable position in the car. We could have incorporated a very slight taper design, but the general shape of the frame had to change. Once we began designing the new frame, we decided we wanted a larger torque output due to the friction being placed on the shaft from the frame. This meant we needed a larger gear on the shaft which made it difficult to include the already established frame of the car. The previous model contained all of the gears and components inside of the car. This made it so the new 96 teeth gear could not be contained inside the car since it would clash against the bottom of the frame. We considered moving the larger gear higher up in the car, extending the length of the transmission with a belt connecting the two gears. However to keep the layout as close to what it was, we decided to keep the transmission height and layout the same and just change the frame of the car to include a slight slot for the larger gear so it could move freely the updates slot can be seen in Figure 3. This gear paired with the 10 teeth gear on the motor, gave us a ratio of 9:1. This iteration was what we used during our proof-of-concept check in class as it had a transmission layout that we were comfortable with and had the basic components we wanted to add to our final design.

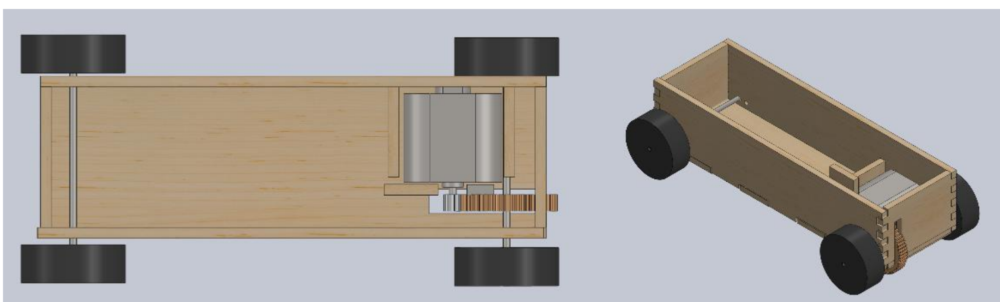


Figure 3: SolidWorks Assembly of updated second iteration

From testing the iteration shown in Figure 3, we gathered that the car was still unable to go 10 feet on its own. We figured it was probably due to the excess weight from a lot of wood being used in the length of the car but not serving any real purpose. We then decided to measure the length and width of the transmission and decided to reduce the area of the bottom of the car to only be able to enclose the transmission and battery holder. We also decided to change the shape of the car. We started with the rectangular frame and decided to slant the front of the car down to reduce some weight and surface area. We also decided to include this to the back of the car but not as steep. The angle for the front cut was 12.92 degrees, and the back cut was 17.73 degrees from the horizontal. This fit our initial goal of reducing the weight of the car so the torque output wouldn't have to much higher due to the light weight of the car. This also made the car slightly more aerodynamic as it reduced the surface area at the front of the car which provided some downforce to the rest of the car. This design overall was a more throughout car that fit more of our initial goals. However, this specific iteration gave us the most trouble for various reasons. The primary one was getting the dimensions for the walls right. Since the top was now at an angle, we couldn't figure out a way to make the cut-out style of putting the pieces together work for all the parts. We attempted to create some on SolidWorks yet when mating the designs together none of them fit accurately and there would be some interference or change in angle for the pieces. Due to this we decided that there would be slight gaps where the two tops meet as well of the cars as well as between the top of the car and the front and back sides. These pieces were still secured onto the car by connecting to the sides of the car which maintained a rectangular shape. The sides were just formed to create a triangular shape to match the new tapered tops angles. This can be seen in Figure 4. We also chose to keep the slight slot for the on and off switch, since the inside of the car would be closed off so it would make it easier to turn on and off without having to deal with the battery holder. The one major flaw we found with this design is that since the car would be closed off, we would have to completely take the car apart before being able to fix the inside of the car and there would be no way to tell how much damage the frame of the car would sustain from it.

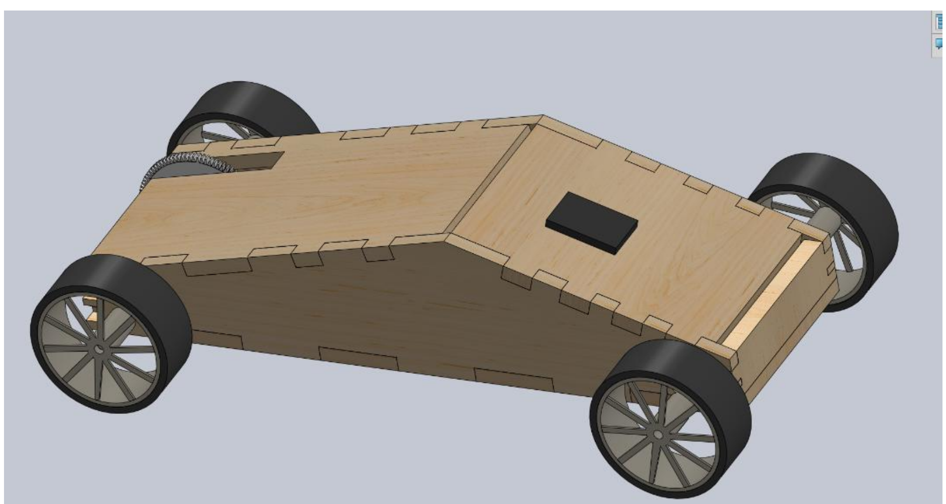


Figure 4: First Solidworks Assembly of our last iteration

We also built one last iteration of this car, in which we tried to make all our original goals happen as seen in Figure 5, with a long narrow nose and front and rear wings. This design kept the slot for the gear and still had the transmission laid out as before however the battery holder would be held in the thin front end. This transmission also would behave like that of an F1 car by being clustered in the back of the car. This car would have a cutout for the motor in which we could glue down the motor. As we had noticed, the motor would move a lot during our previous iteration as it was not contained in a gear box. The hole in the side of the car was meant to reduce some weight by removing some wood while also functioning as a gear box that would hold the motor in place without having to add any extra pieces. This design fit a lot of our original design goals however and looked similar to an F1 car. Yet when put together the cars frame was too heavy and placed too much friction of the shaft so even when the transmission design was the same as the one from the previous iteration, the car was too heavy to move. After this we decided to compromise some aspects of our original goal such as the front and rear wings and the skinny nose for the car, to get the car to fulfill its purpose of traveling 10ft.

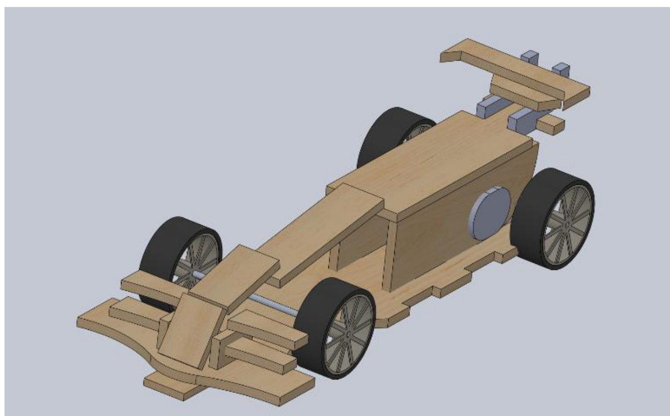


Figure 5: CAD design of our scrapped 4th iteration

This meant that our final design was the one from Figure 4 with two tapered tops as it had gone through the most improvement and fulfilled its purpose. Our team re-cut out the pieces and rebuilt the car however when testing we ran into several issues. The first was that the car would not remain on a linear path we took the car apart to get a better understanding of if the frame itself was crooked but when taking off the shaft and wheels we discovered that the shaft had been bent on the left side. This made the car veer off path, but this was resolved by switching the shafts which resulted in the car going straight. We also measured the speed of the car, and it was unfortunately slower than we anticipated going around 2.3 seconds to complete the ten feet. To fix this issue we decided to only keep on one of the tops of the car and the front side of the car, the one with the switch attached to it. Since we wanted back wheel drive, we had to switch the wires on the motor to switch the direction that the wheels spun as the side with switch on it was originally the back. Therefore, we swapped the front and back of the car, the front was now the side with the 17.73 degree angle and the back was the slant at 12.92 degrees. The car after the changes can be seen in This made it so one of our goals was changed but another was met as the car was no longer as aerodynamic, but it was faster. We also had to use a different motor and restart on the wiring of the car since the wires had been soldering onto the motor and were not able to be removed without leaving residue. Once the inside was set back into place the car was completed and it is seen in Figure 6. We made this specific change of switching the front and back of the car would allow the car to break through the air in a smoother manner by still having one top on rather than having the open side of the car trap air as it moved which would cause drag and slow it down. The switch also made it easy to turn on and off the car without directly messing with the transmission, which is why it was also kept. When tested again the car traveled slightly faster which our group was content with and decided moved fast enough for the overall weight of the car. The final iteration of the car with all of the changes made can be seen in Figure 7 and Figure 8.

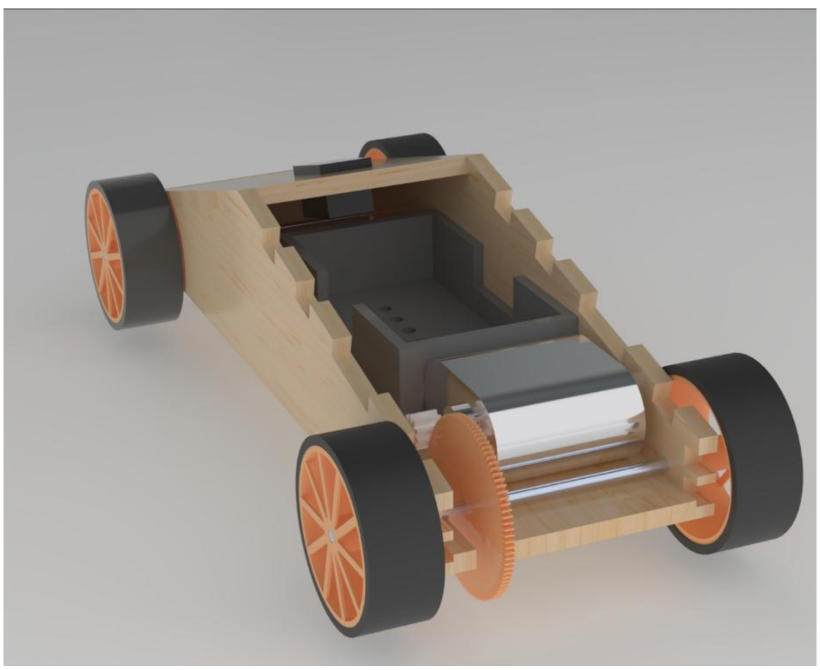
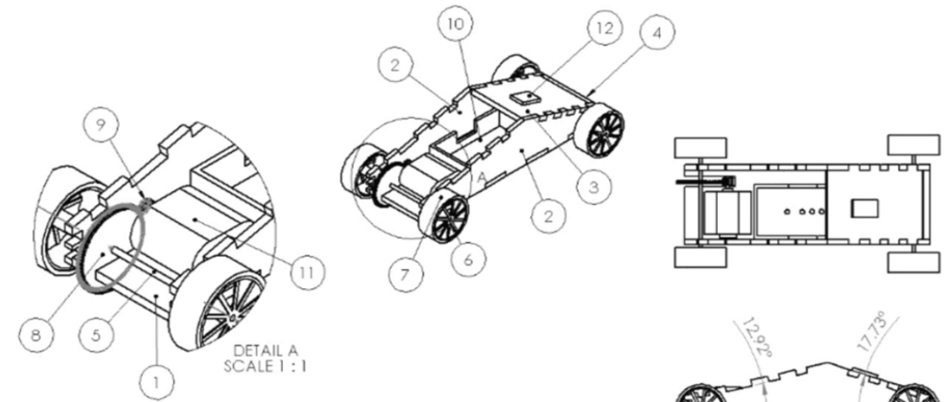
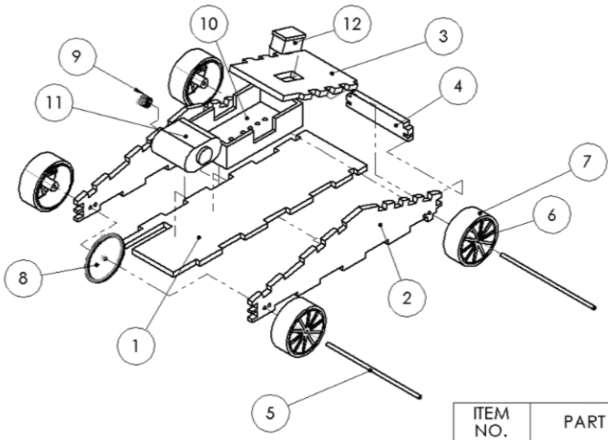


Figure 6:Rendering of Final Design



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	MV_Bottom	1/8" Plywood	1
2	MV_TriangleSide	1/8" Plywood	2
3	MV_FrontTop	1/8" Plywood	1
4	MV_SmallFront	1/8" Plywood	1
5	MV_Shaft	1060 Alloy	2
6	MV_WheelFrame	ABS	4
7	MV_RubberWheel	Rubber	4
8	MV_96ToothGear	ABS	1
9	MV_10ToothGear	ABS	1
10	MV_BatteryHolder	ABS	1
11	TeamMV_motor	Cast Alloy Steel	1
12	MV_Switch	ABS	1

Figure 7: Drawing and Bill of Materials of final iteration



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9	MV_10ToothGear	ABS	1
10	MV_BatteryHolder	ABS	1
11	TeamMV_motor	Cast Alloy Steel	1
12	MV_Switch	ABS	1

Figure 8: Exploded View of final car assembly

There were some issues with the actual putting together of the different iterations of cars. The first issue we ran into was when we attempted to use wood glue to put together the frame. This caused a very messy frame to be put together and made us worry if the glue would get onto the wheel, gears or motor. Therefore, we decided to switch to hot glue which allowed us to have some more control in the way the car was put together and it was easy to peel off if it got on any component, it was not supposed to. The other issue we ran into was during testing of the cars. As we tested its speed, the batteries would heat up a lot in the battery holder and there would be times in which we would see slight smoke. We believe the batteries were too strong. During our final iteration, we ran into the problem that the car was no longer working, when we inspected it we found that the batteries got so hot that they melted through the battery holder and the spring that touched the battery was melted off leaving a hole in the battery holder. The car was once again taken apart and that part replaced, which limited how much actual testing we could do before the final day as we were afraid that the battery holder would once again melt or that the heat from the batteries would impact something else on the car such as the glue or wiring.

Results

Once we fixed several issues and improved the final design to fit the needs of our project and the limitations which we had a successful car. The car traveled the ten feet unassisted in 1.97 seconds. This was faster than the car had been traveling previously and the car traveled along a linear path throughout the entire ten feet. This meant that we managed to create a stable and functional car. The car also weighed 123.4 grams which meant it had a decent weight to speed ratio. We also managed to lower the weight a lot compared to the car in Figure 4 as that car weighed around 146.3 grams.

Although the car did not fit all the goals, we had originally set out to do, we did manage to fit and try out a lot of them throughout the course of the project. Our final design in Figure 7: did not include a front or rear wing but that was mostly due to limitations of the plywood which was required for this project. We also had to compromise the general design of the frame since our final car did not have a long thin nose and featured a slightly different transmission layout than we expected. However, with the resources we had we were able to add some tapers to the top of the car and were able to create a car we were proud of. The wiring all worked perfectly and the switch at the top made it easy to turn on and off as the battery holder would get too hot. Also removing the side and top of the car made it much easier to access the components inside and fix them as small issues became present.

This car also helped to show the trial and error that is required to create a car that fulfills its purpose. Also, it helped to show how important it was to take risks such as we did with the 4th iteration of the car. The car seen in Figure 5 fulfilled mostly all of the aesthetic goals we set out to achieve, yet the car itself was useless. This showed us the tradeoff required to make a useful car and how the aesthetics of a car cannot be prioritized without first prioritizing the functionality of the car. Especially since this project had a very specific purpose of going 10 feet unassisted.

This project also helped us to see how important it is to track progress and keep the plans from previous iterations. If we had not saved the plans from the previous iterations, we would not have had something feasible to show for this project. Overall, this car gave us success because our group was willing to compromise the original plans we had to prioritize the needs of the car. Such as cutting out a slot for the gear and switching the direction of the car to give the car a better chance of improving and fulfilling its purpose. We also learned a lot of skills such as soldering, which improved how easy the car was to turn on and off and taught us some stuff about how cables can be utilized in projects. This car resulted being the best iteration due to the thoughtfulness behind it as the tolerances were considered, the friction on the shaft, weight and aerodynamics were all considered. It took a lot of trial and error, but this car was able to combine what we considered most important to the project and us to give us a proper result.

Conclusions

Creating this car, seemed a lot more simple than it actually ended up being. There required meticulous planning and willingness to go back to the beginning and rework our way around problems in order to get the car to function. Our team found many shortcomings in the beginning of the project which had to be fixed via time commitment to the car and a commitment to learning about cars. When we walked into class the first day of the project and were tasked with creating a transmission, we were dumbfounded. Since then we have grown a deeper understanding of transmissions and all the components that go into a working car.

This project taught us a lot about what being an engineer means and entails. Going from classroom lessons to actual hands-on learning was fun but we saw how necessary the classroom component was. We found ourselves combining our knowledge of physics to understand what limitations the car would face and what improvements could be made. Seeing topics, we hadn’t really considered, such as friction make such a detrimental impact on our car, really helped put into perspective all the knowledge that is needed to be a successful engineer. As well as how willing to try new things engineers are, we were afraid to go out of comfort zone regarding the car, which is why the car remained a rectangle for so long, yet when we finally did is when we started to see some improvement on the car and its performance. We also had to learn not to take failure too deeply, as our failure in the fourth iteration of the car was difficult to stomach as we were quite excited about the cars aesthetics, yet the failure taught us a lot for next time

We were also slightly afraid of the pure freedom we had during this project, as we had to fill in our knowledge gaps and build upon them to make something real. For the first time we fully combined creativity and our engineering skills. Through this we learned that creativity and engineering cannot exist without one another, as when we combined the two, we got a car that we could continue to build off of and improve given time and other resources.

Overall, this project helped show us why we choose to be engineers in the first place, as we have got to create something real and something completely of ours. We saw how we were pushed and why striving to learn will continue to help us now and, in the future, as fully fledged engineers. Working with a very specific set of circumstances also helped show how future projects in our jobs might be as we have budget, time or material limitations. This project was hard and fun and we learned a great deal which we will carry on with us into other classes and our careers.

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